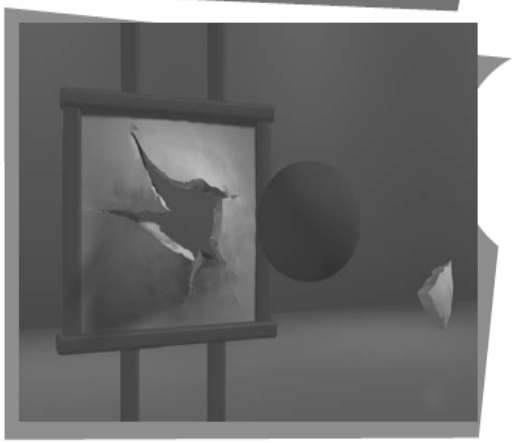
Qassim University
College of computer
جامعة القصيم
كلية الحاسب

COLOR MODELS

Color Models and Color Applications

CHAPTER 17

1 Properties of Light
2 Color Models
3 Standard Primaries and the Chromaticity Diagram
4 The RGB Color Model
5 The YIQ and Related Color Models
6 The CMY and CMYK Color Models
7 The HSV Color Model
8 The HLS Color Model
9 Color Selection and Applications
10 Summary



Our discussions of color up to this point have concentrated on methods involving red, green, and blue (RGB) components, which we use for generating displays on video monitors. Several other color descriptions are useful as well in computer-graphics applications. Some methods are used to describe color output on printers and plotters, some are used for transmitting and storing color information, and others are used to provide a more intuitive color-parameter interface to a program.

Overview - How can we describe and represent colors?

- Properties of light-
- Color
- Color Models – Overview
- CIE
- RGB (Red Green Blue)
- CMY (Cyan Magenta Yellow)
- YIQ
- HSV (Hue Saturation Value)
- HLS (Hue Luminance Saturation) Model

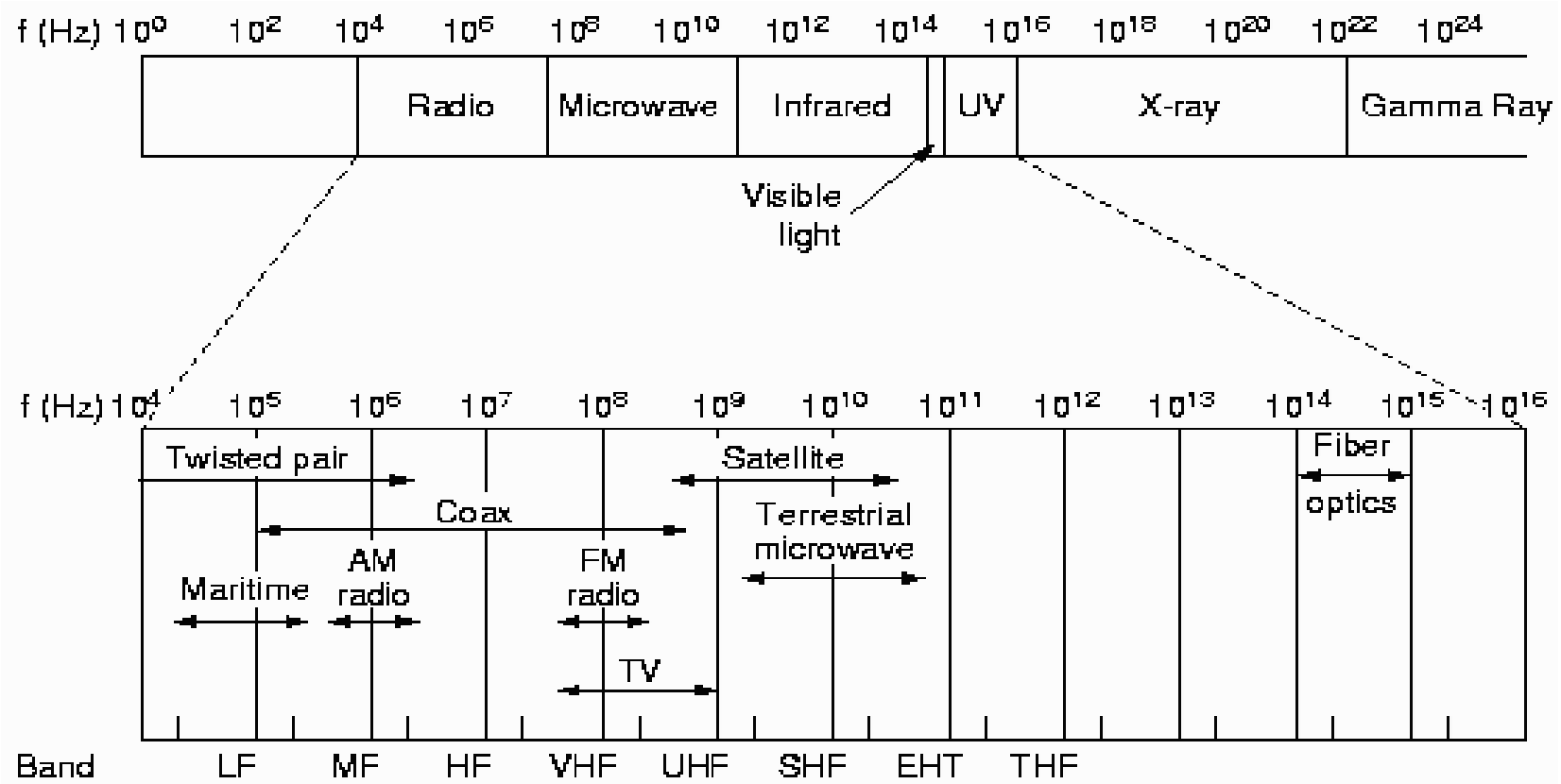
- Up to this point we have used RGB color method for describing color in our discussions.
- Several other color descriptions are also useful in computer graphics applications.
- These color description methods are used for describing color output on display, printing and plotting devices.
- Some are used for transmitting and storing color information.

Physical properties of light

- In physical terms, color is electromagnatic radiations within a narrow frequency band.
- Visible light is part of the electromagnetic radiation (380-750 nm)
- 1 nm (nanometer) = 10^{-10} m (=10^{-7} cm)
- 1 Å (angstrom) = 10 nm
- Radiation can be expressed in wavelength (λ) or frequency (f), $c=\lambda f$, where $c=3 \cdot 10^{10}$ cm/sec

Colors

- Colors
  - A narrow frequency band within the electromagnetic spectrum



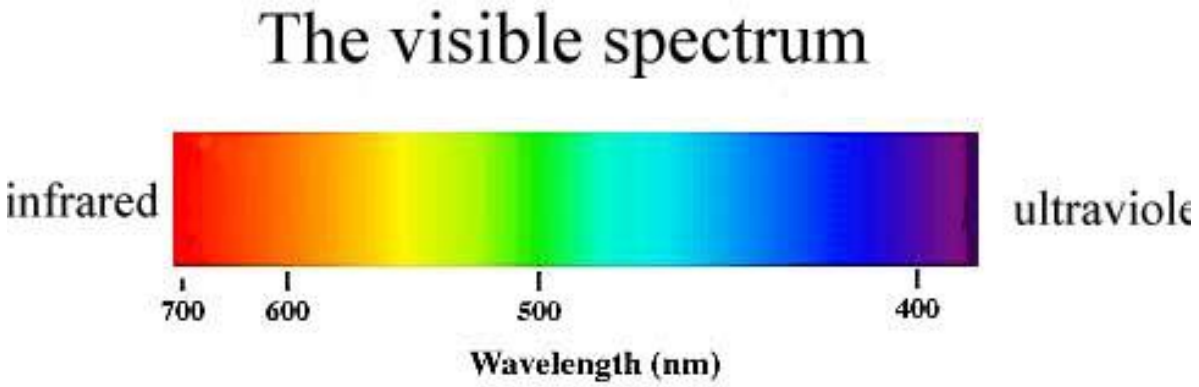
Colors

- **Visible band**
  - Each frequency corresponds to a distinct color
  - Low-frequency end (4.3×10^{14} Hz): Red
  - High-frequency end (7.5×10^{14} Hz): Violet
  - Wavelength $\lambda = v/f$, where v=300,000km/sec
  - Low frequency High frequency

red orange yellow green blue violet

Long wavelength Short wavelength

700nm 400nm



Psychological properties of light

When we view a source of light, our eyes respond to:

1-Color (Dominant frequency) of the light.
2-Brightness of light: which corresponds to the total light energy and can be quantified as the luminance of the light.
3-Purity or Saturation of light: Purity describes how close a light appears to be to a pure spectral color, such as red
4-Chromaticity - term is used to describe collectively § (hue).
5- Luminance: The quality of being luminous; emitting or reflecting light

Color definitions

Complementary colors - two colors combine to produce white light

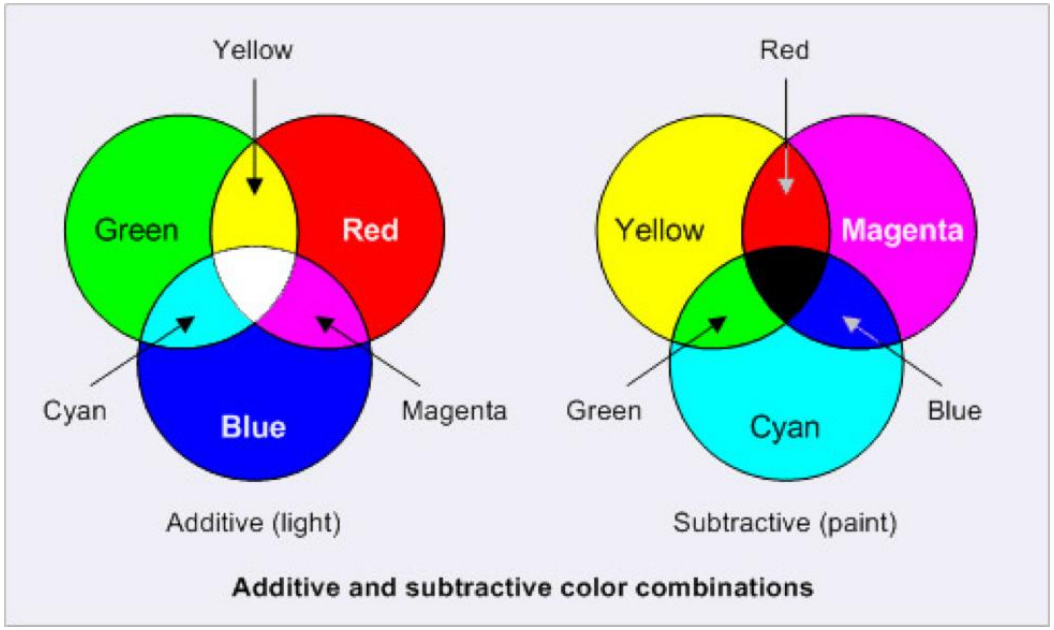
Primary colors - (two or) three colors used for describing other colors

Two main principles for mixing colors:

1. *additive mixing*
2. *subtractive mixing*

Additive Mixing

- Pure colors are put close to each other => a mix on the retina of the human eye (cp. RGB)
- Overlapping gives yellow, cyan, magenta and white
- The typical technique on color displays



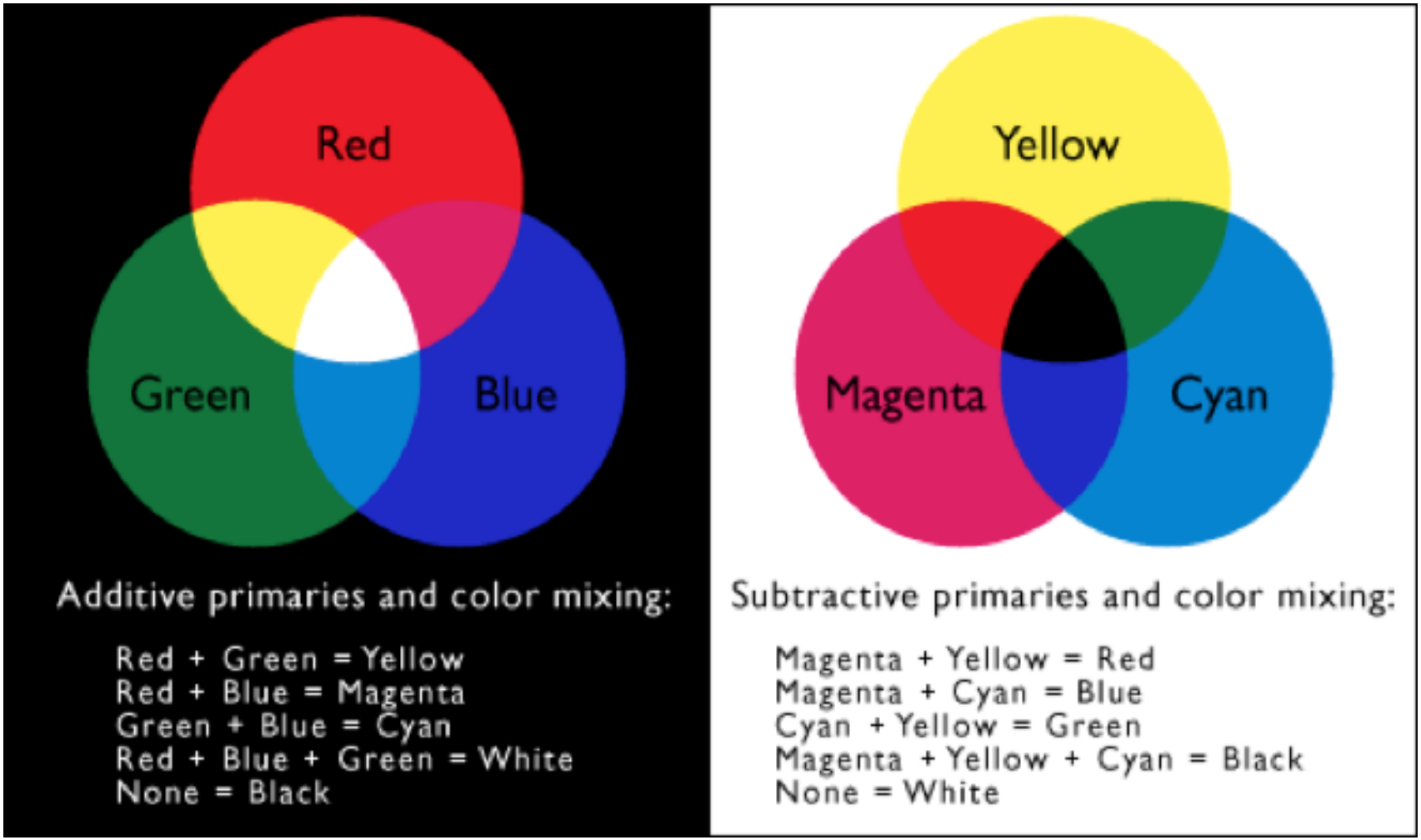
Subtractive Mixing

- Color pigments are mixed directly in some liquid, e.g. ink
- Each color in the mixture absorbs its specific part of the incident light
- The color of the mixture is determined by subtraction of colored light, e.g. yellow absorbs blue => only red and green, i.e. yellow, will reach the eye (yellow because of addition)

Subtractive Mixing

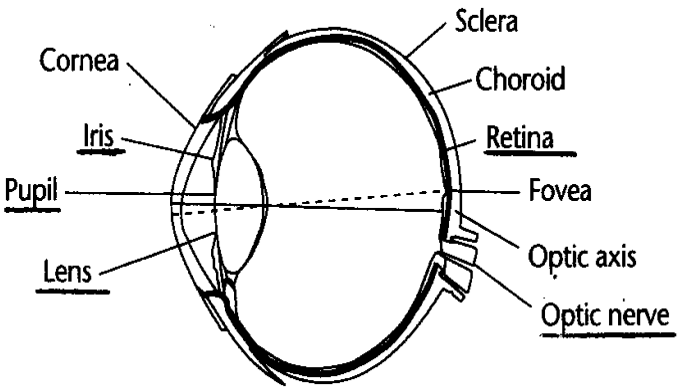
- Primary colors: cyan, magenta and yellow, i.e. CMY
- The typical technique in printers/plotters
- Connection between additive and subtractive primary colors (cp. the color models RGB and CMY)

Additive/subtractive Mixing

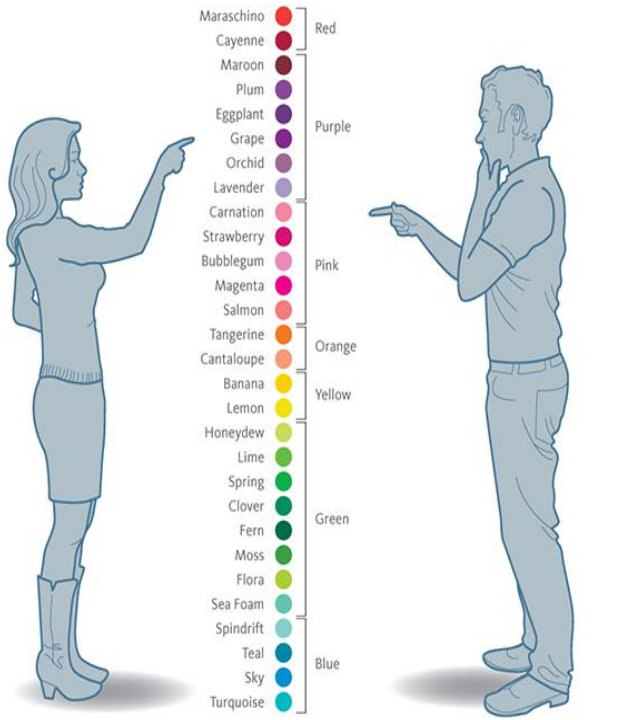


Human Color Seeing

The retina of the human eye consists of *cones* (7-8M),"tappar", and *rods* (100-120M), "stavar", which are connected with nerve fibres to the brain.



1 Structures of the human eye. (Adapted from Atkinson.[2])



Human color seeing

-Theory: the cones consist of various light absorbing material

-The light sensitivity of the cones and rods varies with the wavelength, and between persons

-The ”sum” of

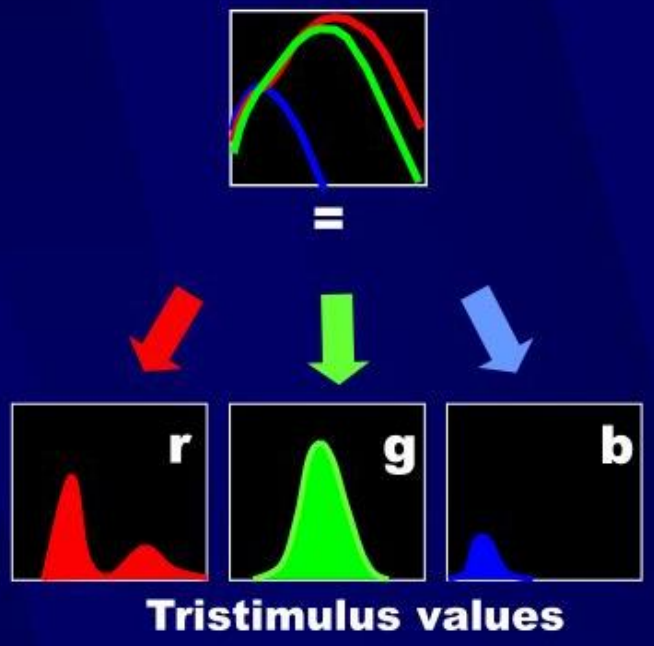
- The energy spectrum of the light
- The reflection spectrum of the object
- The response spectrum of the eye

decides the color perception for a person

Theory of Trichromatic Vision

- The principle that the color you see depends on signals from the three types of cones (L, M, S).
- The principle that visible color can be mapped in terms of the three colors (R, G, B) is called ***trichromacy***.
- The three numbers used to represent the different intensities of red, green, and blue needed are called ***tristimulus values***.





Color Models

- Any method for explaining the properties or behavior of color within some particular context is called a color model.
- No single model can explain all aspects of color, so we will discuss different models here.

- **Primary colors:** basic color (hue) used to produce other colors.
- **Color gamut:** set of all colors that we can produce from mixing the primary colors.
- **Complementary color:** two primary colors that produce whit color.
- **Shades** (different) of color: are produced by adding a black pigment to the primary colors or pure colors or pure hue.
- **Tints** of the color: are obtained by adding a white pigment to the original color, making it lighter.
- **Tones** of color: are produced by adding both black and white pigments to the pigments to the pure color.

Overview of color models

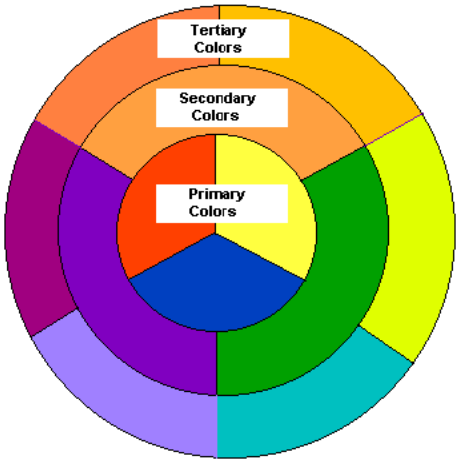
The human eye can perceive about 382000(!) different colors

Necessary with some kind of classification system; all using three coordinates as a basis:

1) CIE standard.
2) RGB color Model.
3) CMY color Model (also, CMYK).
4) YIQ.
5) HSV color Model.
6) HLS color Model.

CIE Color Model

- Color models
  - Use three primary colors to produce other colors
- Primary colors
  - Colors used in a color model to produce all the other colors in that model.
  - Cannot be made from the other (two) colors defining the model.
- CIE color model (International Commission on Illumination – 1931)
  - *X*, *Y*, and *Z*: nonexistent, super saturated Colors.
    - Vectors in 3-D additive color space
  - Any color $S = AX + BY + CZ$

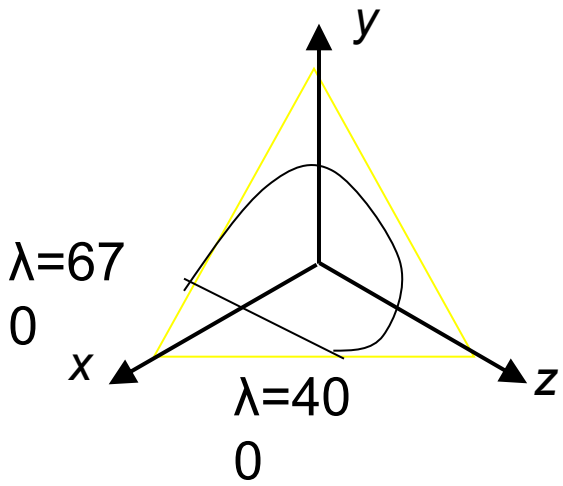


CIE Color Model

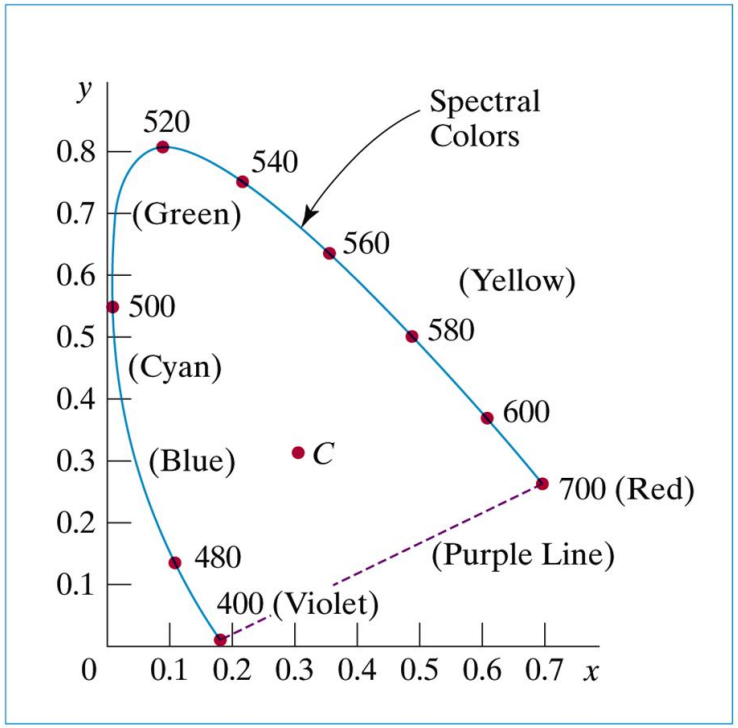
- $S = AX + BY + CZ$ can be normalized to
  - $x = A/(A+B+C)$
  - $y = B/(A+B+C)$
  - $z = C/(A+B+C)$

  ➔ **s** $= xX + y\,Y + zZ$, where $x + y + z = 1$

  ➔ **s** lies in the plane $x + y + z = 1$ in 3D



CIE Chromaticity Diagram



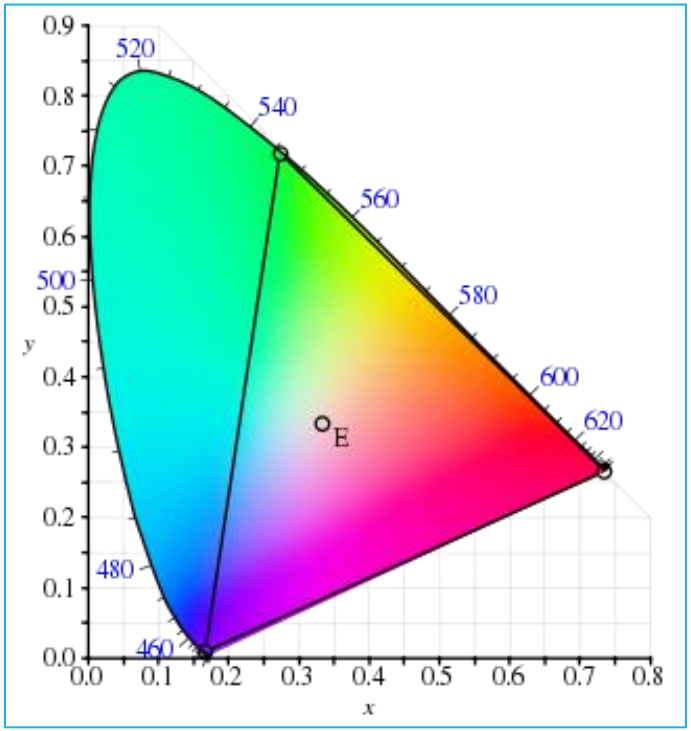


Figure 12-7

CIE chromaticity diagram for the spectral colors from 400 nm to 700 nm.

- When we plot the normalized amounts of x and y in the visible spectrum, we obtain tongue shaped curve – called **CIE chromaticity diagram.**
- Points along the curve are *spectral colors* (pure colors).
- Doted line joining red & violet spectral points is *purple line,* & is not part of the spectrum.
- Interior points represent all possible visible color combinations.
- Point C corresponds to white-light position.
- Luminance values not available in chromaticity (*x, y*) diagram because of normalization.
- Chromaticity diagram is useful for:
  1. Comparing color gamuts for different set of primaries.
  2. Identifying complementary colors.
  3. Determining purity and dominant wavelength for a given color.

Color gamuts

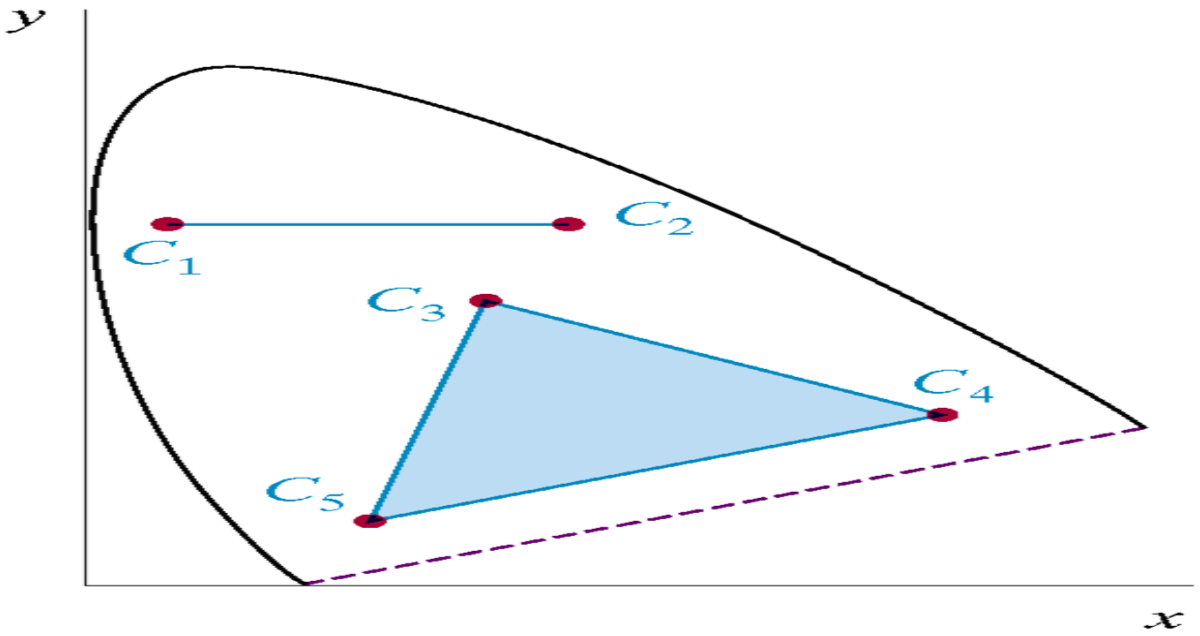


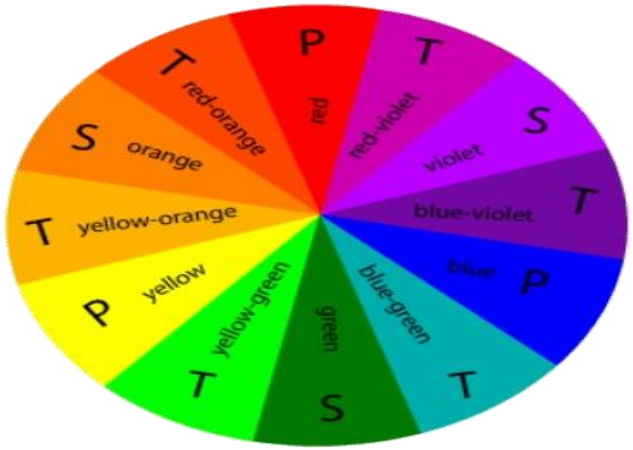
Figure 12-8

Color gamuts defined on the chromaticity diagram for a two-color and a three-color system of primaries.

Color gamuts

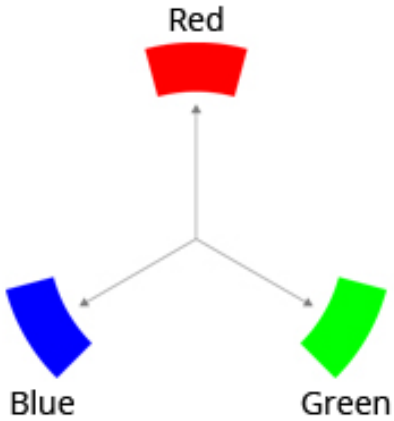
- All colors along straight line joining the points C_1 and C_2 can be obtained by mixing appropriate amounts of the colors C_1 and C_2. If greater proportion of C_1 is used, the resultant color is closer to C1 and vice versa.
- The color gamut for three points, such as C_3, C_4 and C_5 is a triangle with vertices at the three color positions. These three primaries can generate only the colors inside or on the bounding edges of the triangle.
- Color gamuts for video monitors and hard-copy devices are conveniently compared on the chromaticity diagram.

Complementary colours

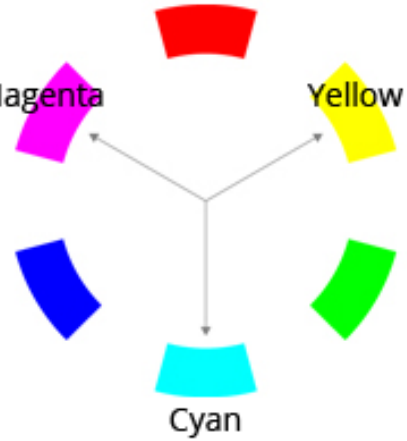


Complementary colours are any two colours which are directly opposite each other, such as red and green and red-violet and yellow-green.

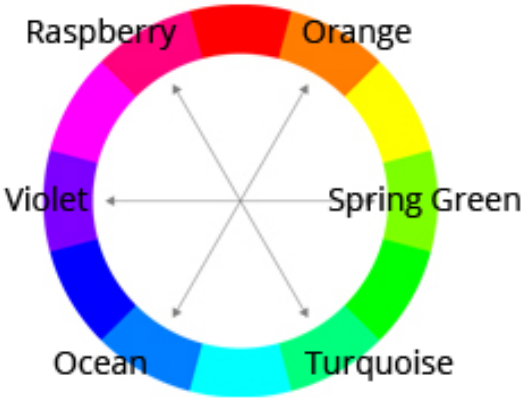
Primary



Secondary



Tertiary



Complementary colors

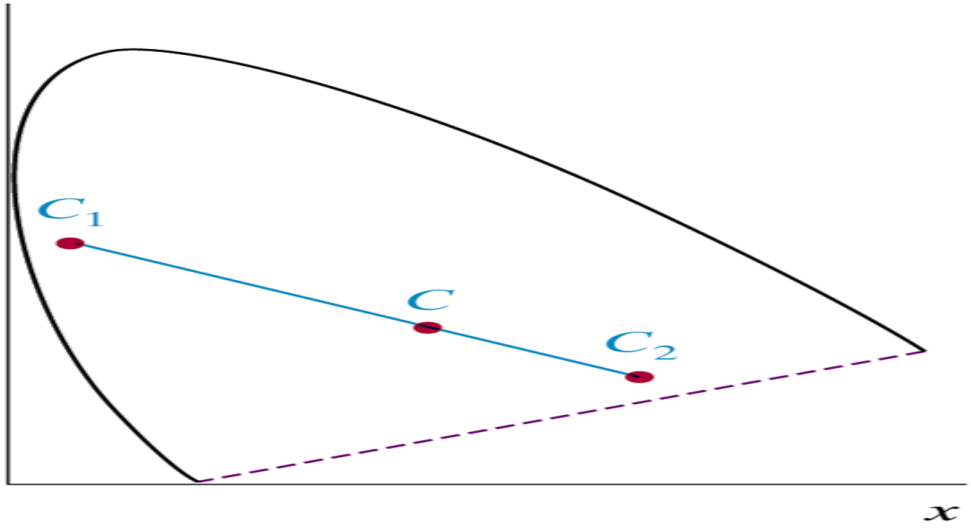


Figure 12-9
Representing complementary colors on the chromaticity diagram.

Complementary colors must be represented on the chromaticity diagram as two points on opposite sides of C and collinear with C. The distances of the two colors C_1 and C_2 to C determine the amounts of each needed to produced white light.

Dominant wavelength

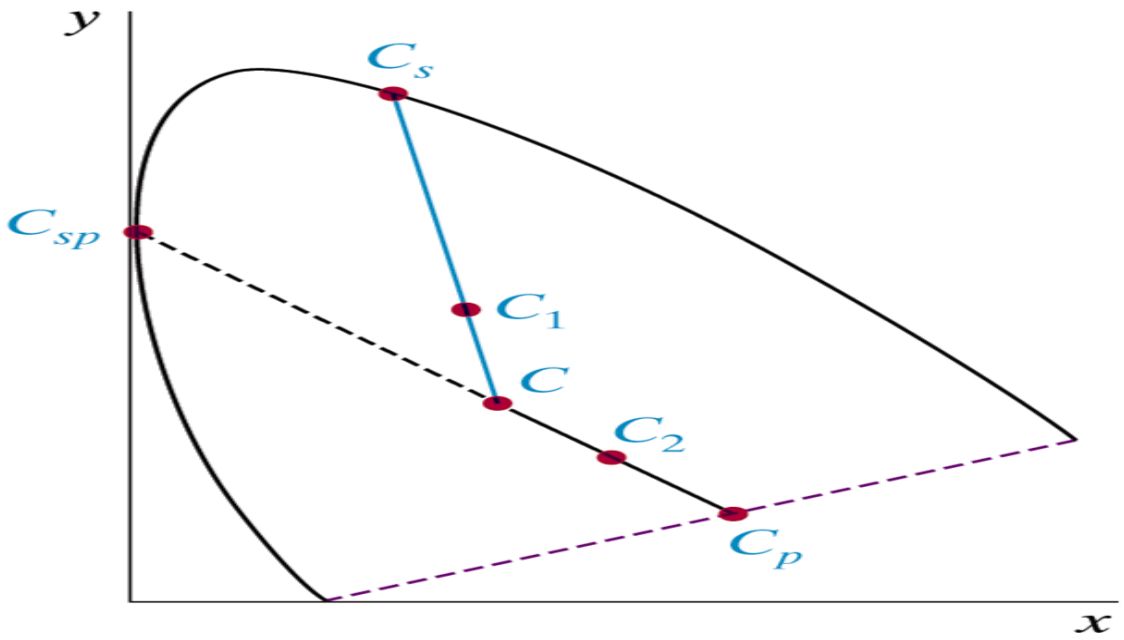


Figure 12-10

Determining dominant wavelength and purity using the chromaticity diagram.

Dominant wavelength

- To determine the dominant wavelength of a color, we draw a straight line from C through that color point to a spectral color on the chromaticity curve. The spectral color C_S is the dominant wavelength for the color C_1. Thus, color C_1 can be represented as a combination of white light C and the spectral color C_S.
- Above method for determining dominant wavelength will not work for color points that are between C & the purple line.
- Drawing a line from C through point C_2 takes us to point C_p on the purple line, which is not in the visible spectrum. In this case we take the compliment of C_p on the spectral curve , which is C_{sp}, as the dominant wavelength.
- Colors such as C_2 have spectral distribution with subtractive dominant wavelengths. We can describe such colors by subtracting the spectral dominant wavelength from white light.

Purity

- In fig. 12.10, we determine the purity as the relative distance of C_1 from C along the straight line joining C to C_s.
- Purity is ratio d_{c1}/d_{cs}
- Where
  - d_{c1} = distance from C to C_1
  - d_{cs} = distance from C to C_s

Uses of CIE Chromaticity Diagram

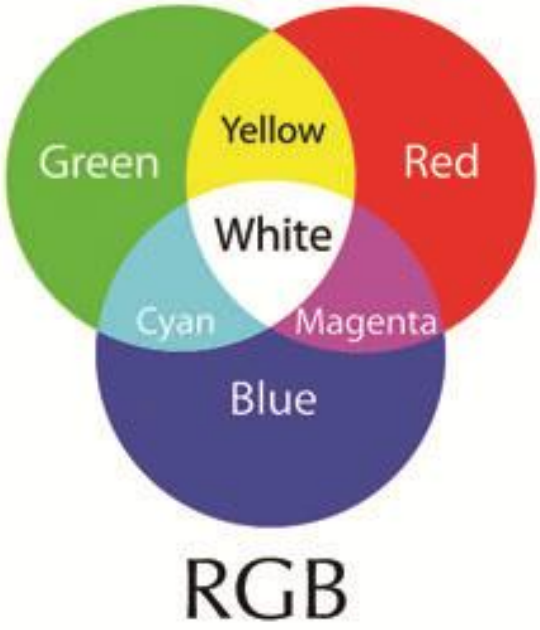
- Define *color gamuts*
  - Range of colors that can be produced on a device
- CRT monitor's gamut is different from printer's
- Any choice of three primaries can never encompass all visible colors
- RGB are natural choices for primaries as they can cover the largest part of the "horseshoe"

RGB model

- Electronics displays use the RGB model, which means that the colors are not absolute, but rather dependent upon the sensitivities and settings of individual devices.
- Used in CRT, LCD, plasma, & LED displays.
- A video monitor displays color patterns by combining light that is emitted from the screen phosphors, which is an additive process. Emitting devices – surface is Black => Origin=> Black, (0, 0, 0).

RGB Color Model

- RGB color scheme is an Additive model:
  - Result = individual contributions of each primary color added together
  - $C = rR + gG + bB$, where r, g, $b \in [0, 1]$
  - $R = (1, 0, 0)$, $G = (0, 1, 0)$, $B = (0, 0, 1)$

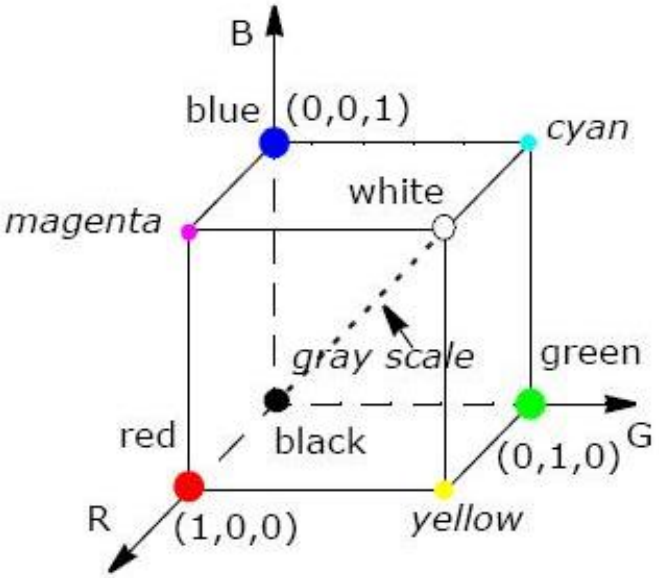


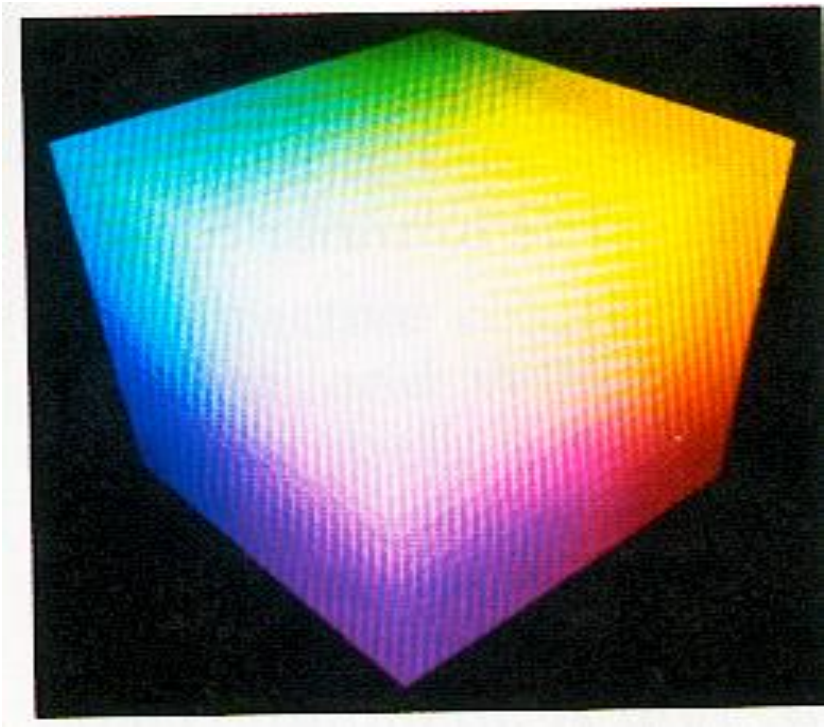
RGB Color Model

- Color Cube
  - $R + G = (1, 0, 0) + (0, 1, 0) = (1, 1, 0) = Y$
  - $R + B = (1, 0, 0) + (0, 0, 1) = (1, 0, 1) = M$
  - $B + G = (0, 0, 1) + (0, 1, 0) = (0, 1, 1) = C$
  - $R + G + B = (1, 1, 1) = W$
  - $1 - W = (0, 0, 0) = \text{BLK}$
  - Grays = (x, x, x), where $x \in (0, 1)$, $(x = 0.5)$

RGB Color Cube

- The *RGB cube*
- 1 bit/primary => 8 colors, 8 bits/primary => 16M colors





Origin=> **Black** (0, 0, 0) & White(1,1,1)

CMY/CMYK Color Model

- CMY: Complements of RGB – Subtractive
- Used in light absorbing devices
  - Hardcopy output devices – Printers, plotters
- Hardcopy devices produce a color picture by coating a paper with color pigments. We see the color patterns on the paper by reflected light, which is subtractive process. Absorbing devices – surface is White => Origin=> White.
- Subtractive
  - Color specified by what is subtracted from white light
  - Cyan absorbs red, magenta absorbs green, and yellow absorbs blue.

CMY Color Model

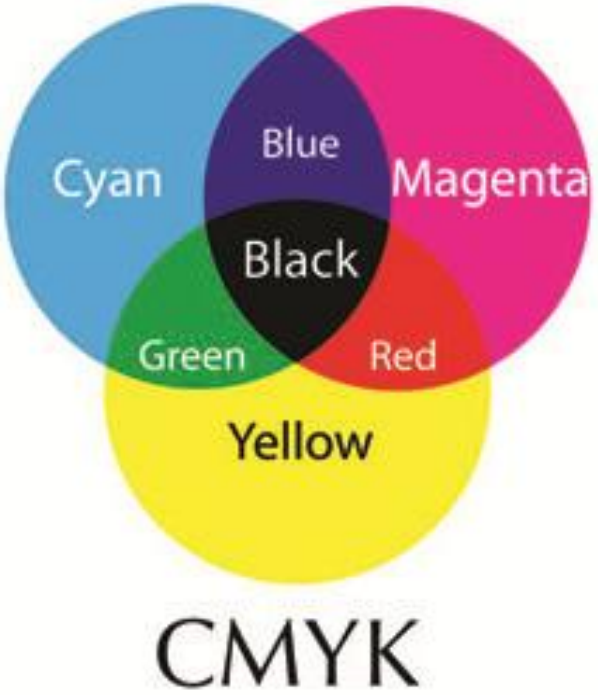
- Unlike RGB, which is an additive color model, CMYK is a subtractive color model.

- Typically used in printing, CMYK assumes that the background is white, and thus subtracts the assumed brightness of the white background from four colors: cyan, magenta, yellow, and black (called “key”).
(where K is the black parameter)

- **Black** is used because the combination of the three primary colors (CMY) doesn’t produce a fully saturated black.

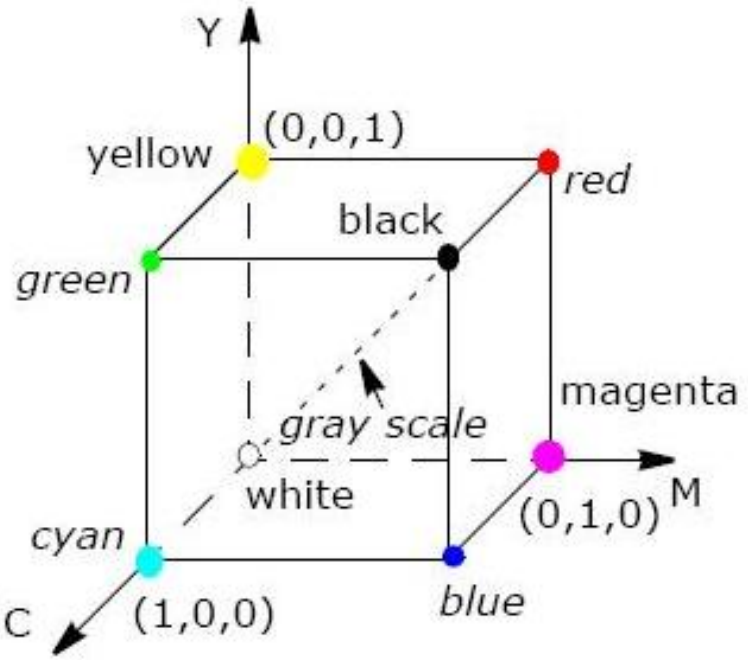
CMY Color Model

- Cyan, Magenta and Yellow are comple-mentary colors of red, green and blue respectively
- Subtractive mixing
- The typical printer technique



CMY Color Model

- Almost the same cube as with RGB; only inverse the places of black and white.
- The various colors are obtained by reducing light.
- E.g. When white light is refletced from cyan colored ink, the reflected light contains only the green & blue components, and red component is absorbed or subtracted, by the ink.



Origin=> **White** (0,0,0)
& Black(1,1,1)

RGB vs CMY

If the intensities are represented as $0 \le r,g,b \le 1$ and $0 \le c,m,y \le 1$ (also coordinates 0-255 can be used), then the relation between RGB and CMY can be described as:

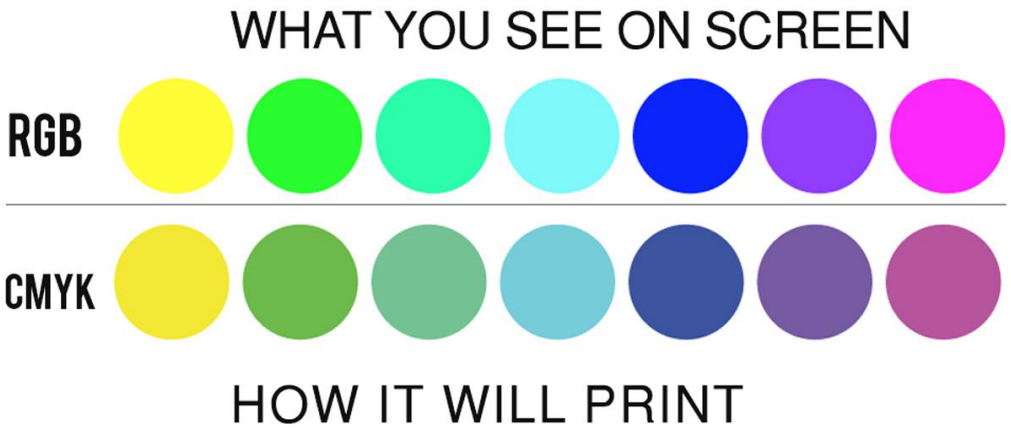
- $W = (0, 0, 0)$ $B = (1, 1, 1)$
- Conversion from RGB to CMY
- Unit column vector represents white point in RGB space

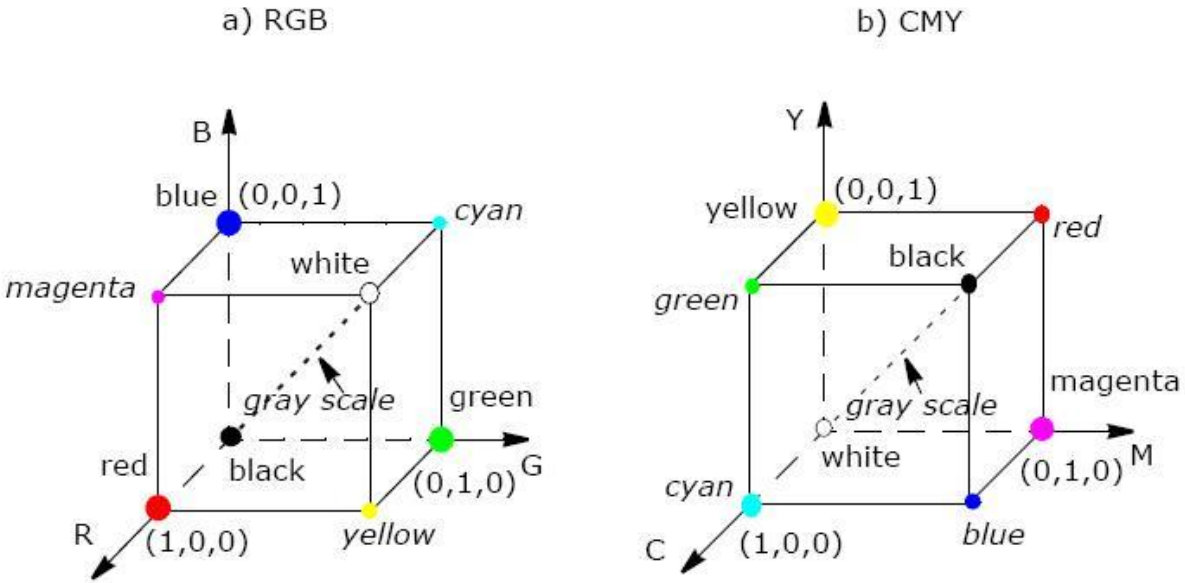
$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- Conversion from CMY to RGB
- Unit column vector represents black point in CMY space.

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix}$$

RGB vs CMY


WHAT YOU SEE ON SCREEN
RGB
CMYK
HOW IT WILL PRINT


a) RGB
B
blue
(0,0,1)
cyan
white
magenta
gray scale
green
red
black
(0,1,0)
G
R
(1,0,0)
yellow
b) CMY
Y
yellow
(0,0,1)
red
black
green
magenta
gray scale
white
(0,1,0)
M
cyan
C
(1,0,0)
blue

CMYK Color Model

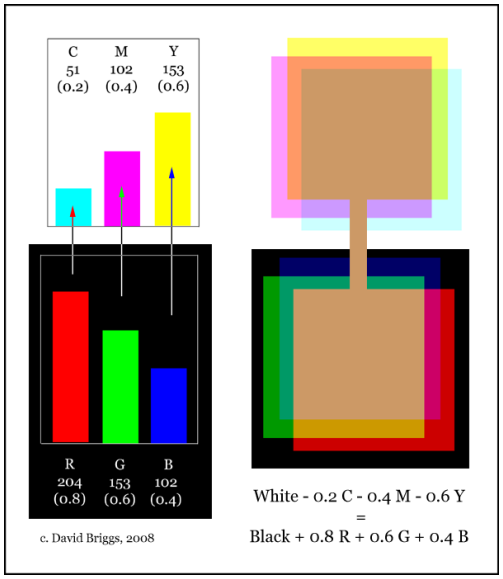
- For printing and graphics art industry, CMY is not enough; a fourth primary, K which stands for black, is added.
- Conversions between RGB and CMYK are possible, although they require some extra processing.
- **For the transformation from RGB to CMYK**
  - First set K = *max*(R,G,B)
  - Then K is subtracted from each of C,M & Y.
- **Similarly, for the transformation from CMYK to RGB**
  - First set K = *min*(R, G, B)
  - Then K is subtracted from each of R, G & B.

YIQ Color Model

- RGB monitor requires separate signals for the red, green and blue components of an image, a television monitor uses a composite signal.
- NTSC color encoding for forming the composite video signal is called the YIQ model.
- Used in U.S. commercial color-TV broadcasting
  - Recoding of RGB for transmission efficiency
  - Backward compatible with black-and-white TV
  - Transmitted using NTSC (National Television System Committee) standard

YIQ Color Model

- YIQ
  - *Y*: luminance (brightness) information
  - *I*, *Q*: chromaticity (hue & purity) information
  - Only *Y* signal is used by black-and-white television monitors
- **Transformation matrix for conversion from RGB space to YIQ space.**



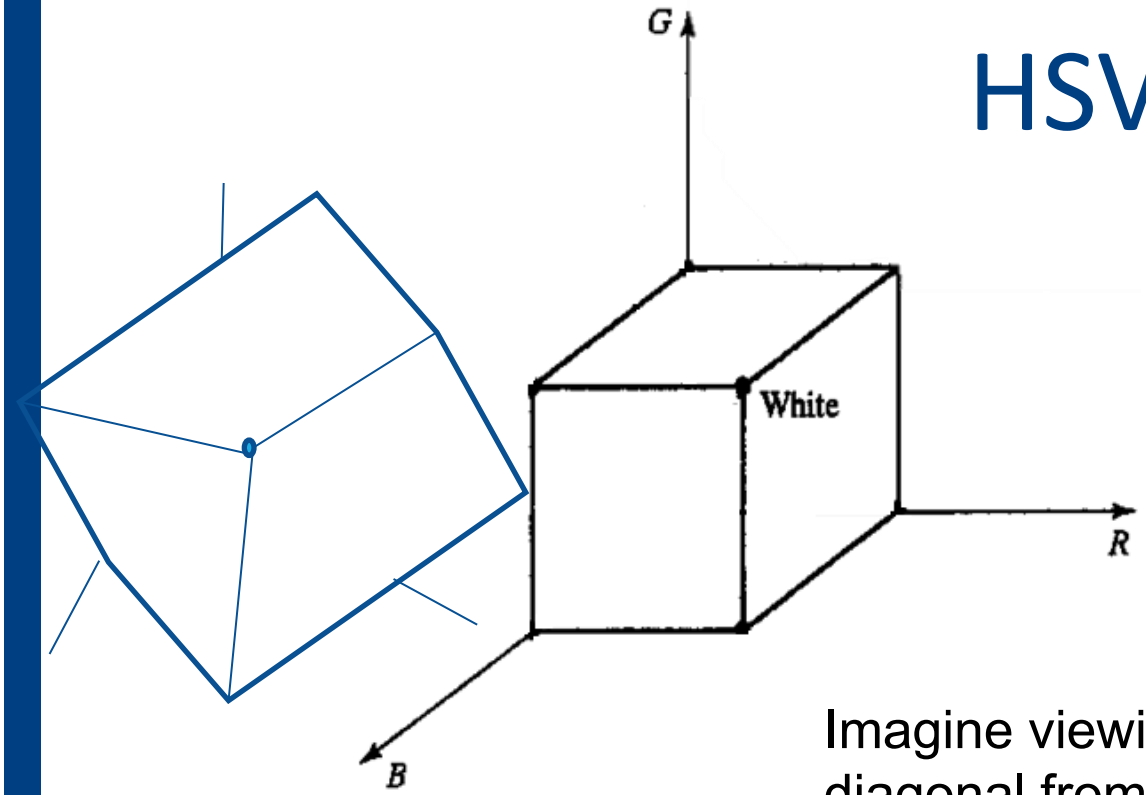
$$\begin{pmatrix} Y \\ I \\ Q \end{pmatrix} = \begin{pmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.275 & -0.321 \\ 0.212 & -0.528 & 0.311 \end{pmatrix} \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

- Y = 0.299 R + 0.587 G + 0.114 B, I = . . .

HSV Color Model

- HSV = Hue, Saturation, and Value are **Color parameters**
  - HSB, where B is Brightness
- RGB, CMY, and YIQ: hardware-oriented
- HSV and HLS: user-oriented
- Cylinder coordinate system
  - Color Space: hex-cone
  - Hexagon is obtained from the color cube in isometric projection.

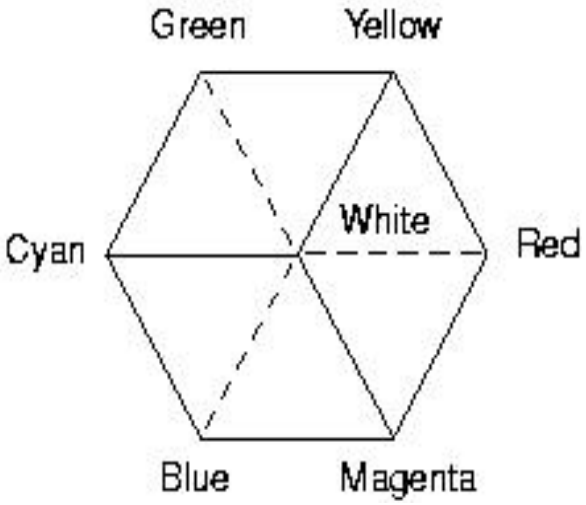
HSV Color Model



Imagine viewing RGB cube along the diagonal from the white vertex to the origin (Black), we see the outline of a hexagon shape. The boundary of hexagon represents various hues, and it is used as the top of the HSV hexagon.

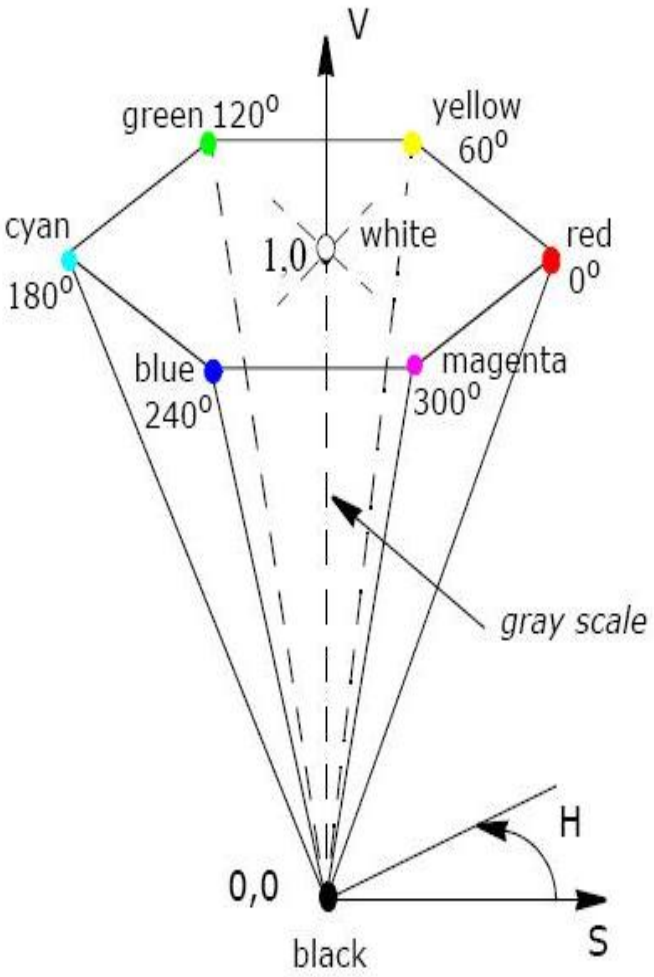
HSV Color Model

Color Cube



-(h, s, v), where h ∈ [0°, 360°) and s, v ∈ [0, 1]
-Hue: angle round the hexagon
-saturation: distance from the center
-value: axis through the center

Hex-cone



HSV Color Model

In HSV space:

- **Hue H:**
  - -The boundary of the hexagon represent various hues (H).
  - -Hue is represented as an angle about the vertical axis, ranging from 0° at red through 360° at magenta.
- **Saturation S:**
  - -Is measured along horizontal axis.
  - -Used to designate the purity of color.
  - -A pure color has the value S=1.0, decreasing S value tend towards the gray-scale.
- **Value parameter V:**
  - -Is measured along a vertical axis through the center of the hex-cone.
  - -V varies from 0 at apex of the hex-cone (Black point) to 1.0 at the top plane (White point).
  - -When V=1.0 and V=1.0, we have the pure hues.

E.g.: To get a dark blue: V=0.4, S=1.0 & H=240°
To get a Light blue: V=1.0, S=0.3 & H=240°

Intuitive Color Concepts

- Tints, shades, and tones ➔ different colors of same hue are produced

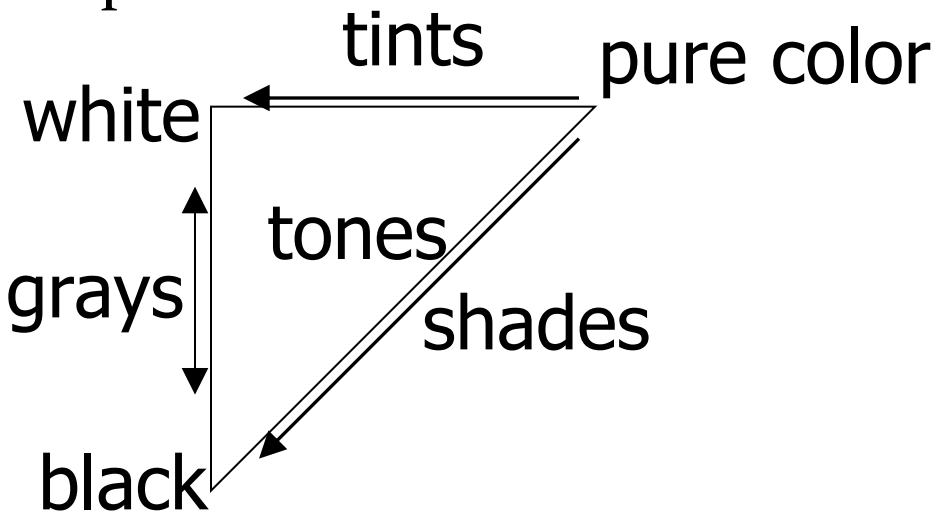
- Grays = black pigments + white pigments



- Graphics packages that provide color palettes to users often employ two or more color models

Intuitive Color Concepts

- Color regions for selecting shades, tints and tones are represented in the cross sectional plane of the HSV hexcone:



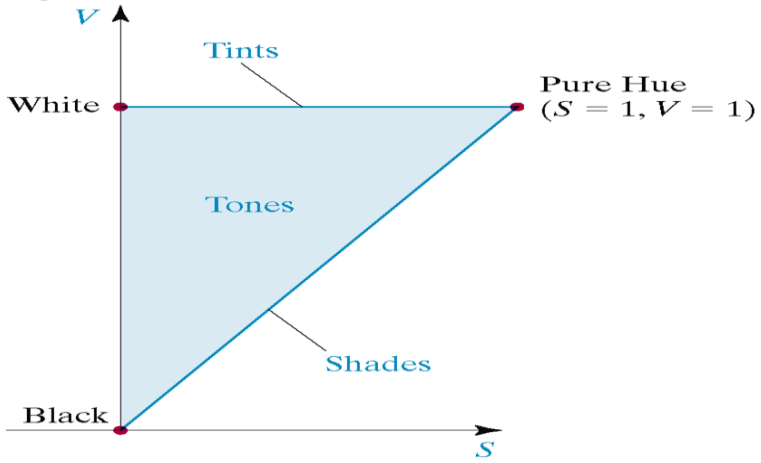


Figure 12-17
Cross section of the HSV hexcone, showing regions for shades, tints, and tones.

Intuitive Color Concepts

-**Tint:** white pigment added to pure pigment,

$V=1,\ 0.0 \leq S \leq 1.0$

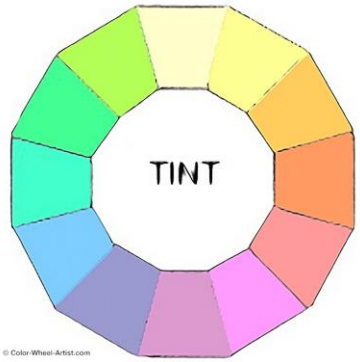
➔ saturation reduced

-**Shade:** black pigment added to pure pigment,

$S=1,\ 0.0 \leq V \leq 1.0$

➔ lightness reduced

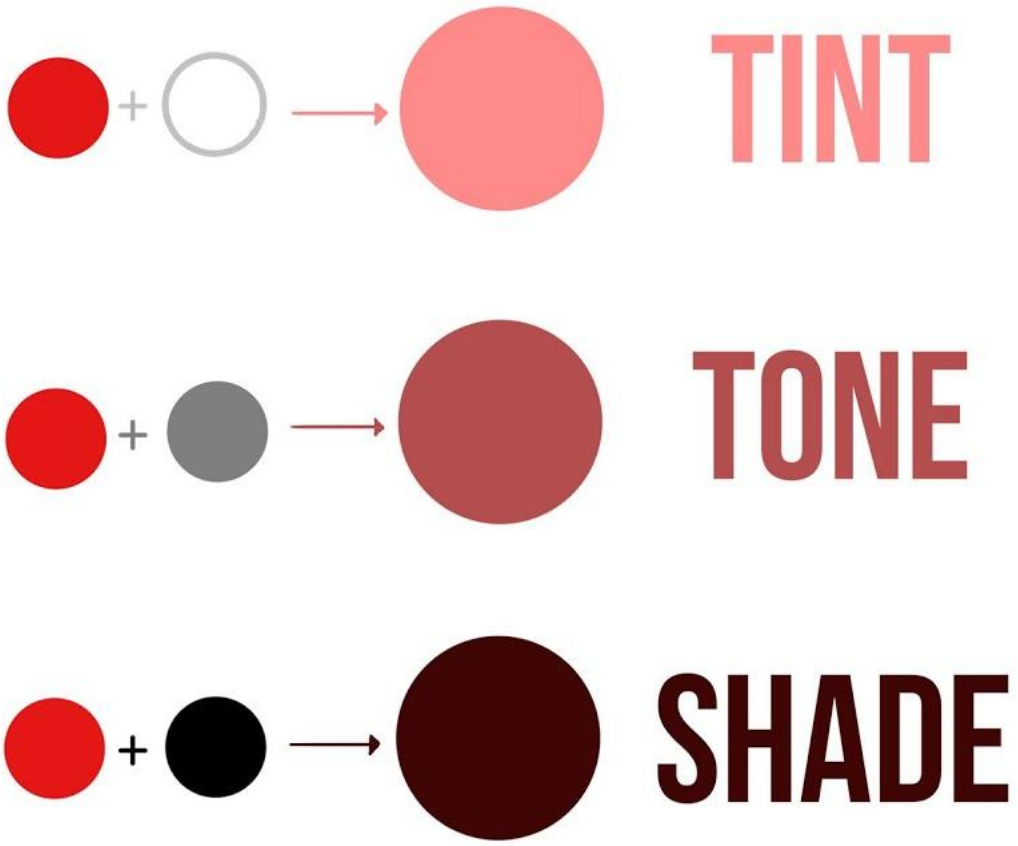
-**Tone:** consequence of adding both white and black pigments to pure pigments, both **saturation & lightness** reduces.







Intuitive Color Concepts



HSV Color Model

- $W = (-, 0, 1)$
- $B = (-, 0, 0)$
- $R = (0, 1, 1)$

$Y = (60, 1, 1)$

:

$M = (300, 1, 1)$

HSV Color Model

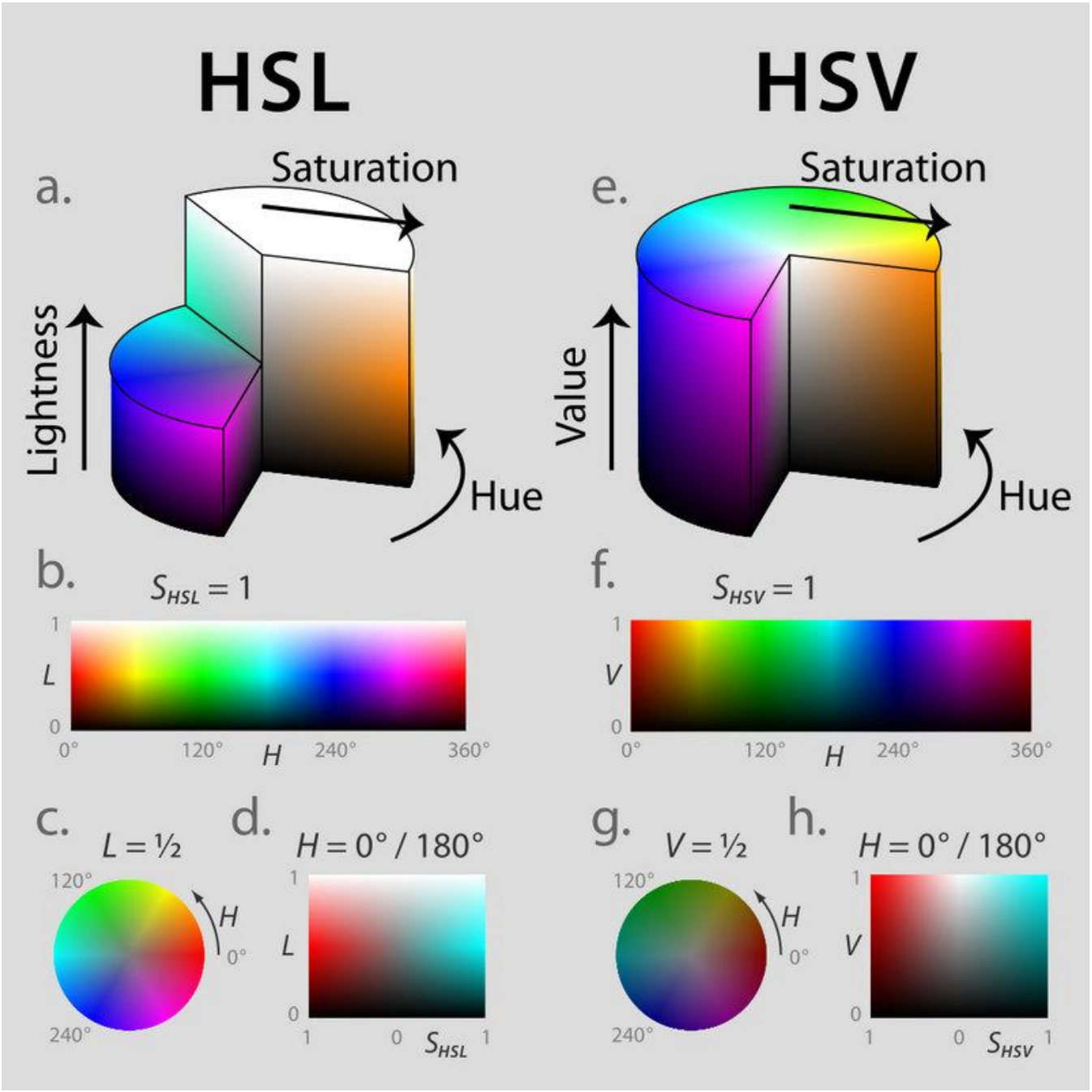
- **True color system:** 16 million colors

- Human eyes can distinguish
  - 128 hues
  - 130 tints (saturation levels)
  - 23 shades of yellow colors, 16 of blue colors

  ➔ 128 x 130 x 23 = 382720 colors

HLS Color Model

- **HLS: Hue, Lightness, and Saturation.**
- Cylinder coordinate system
  - Color Space: double cone
  - base is from the hexagon as in HSV
  - (h, l, s), where h ∈ [0°, 360°) and l, s ∈ [0, 1]
    - **H-hue:** angle around the base
    - **L-lightness:** vertical axis through the center, L=0=Black, L=1=White, grayscale values are along L axis & pure colors lie on L=0.5 plane.
    - **S - saturation:** distance from the center
- *White* = (-, 1, 0)
- *Black* = (-, 0, 0)
- *Red* = (0, 0.5, 1), *Yellow* = (60, 0.5, 1), …


HSL
HSV
a.
Saturation
Lightness
Hue
e.
Saturation
Value
Hue
b.
$S_{HSL} = 1$
1
L
0
0°
120°
H
240°
360°
f.
$S_{HSV} = 1$
1
V
0
0°
120°
H
240°
360°
c.
L = ½
120°
H
0°
240°
d.
H = 0° / 180°
1
L
0
1
0
S_{HSL}
1
g.
V = ½
120°
H
0°
240°
h.
H = 0° / 180°
1
V
0
1
0
S_{HSV}
1

HLS Color Model

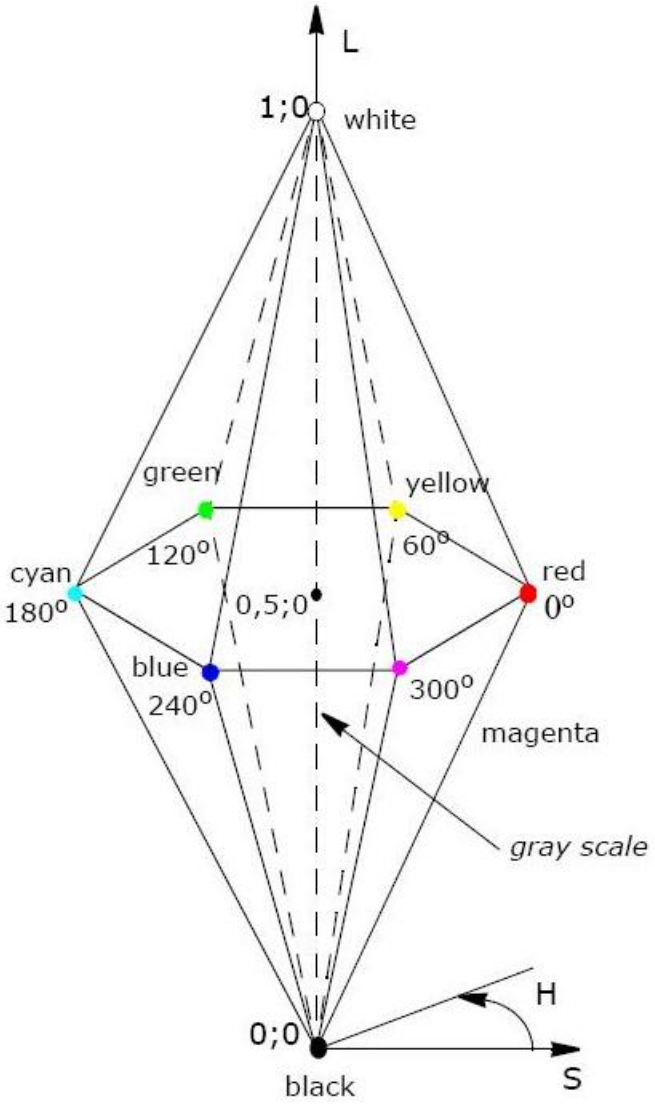
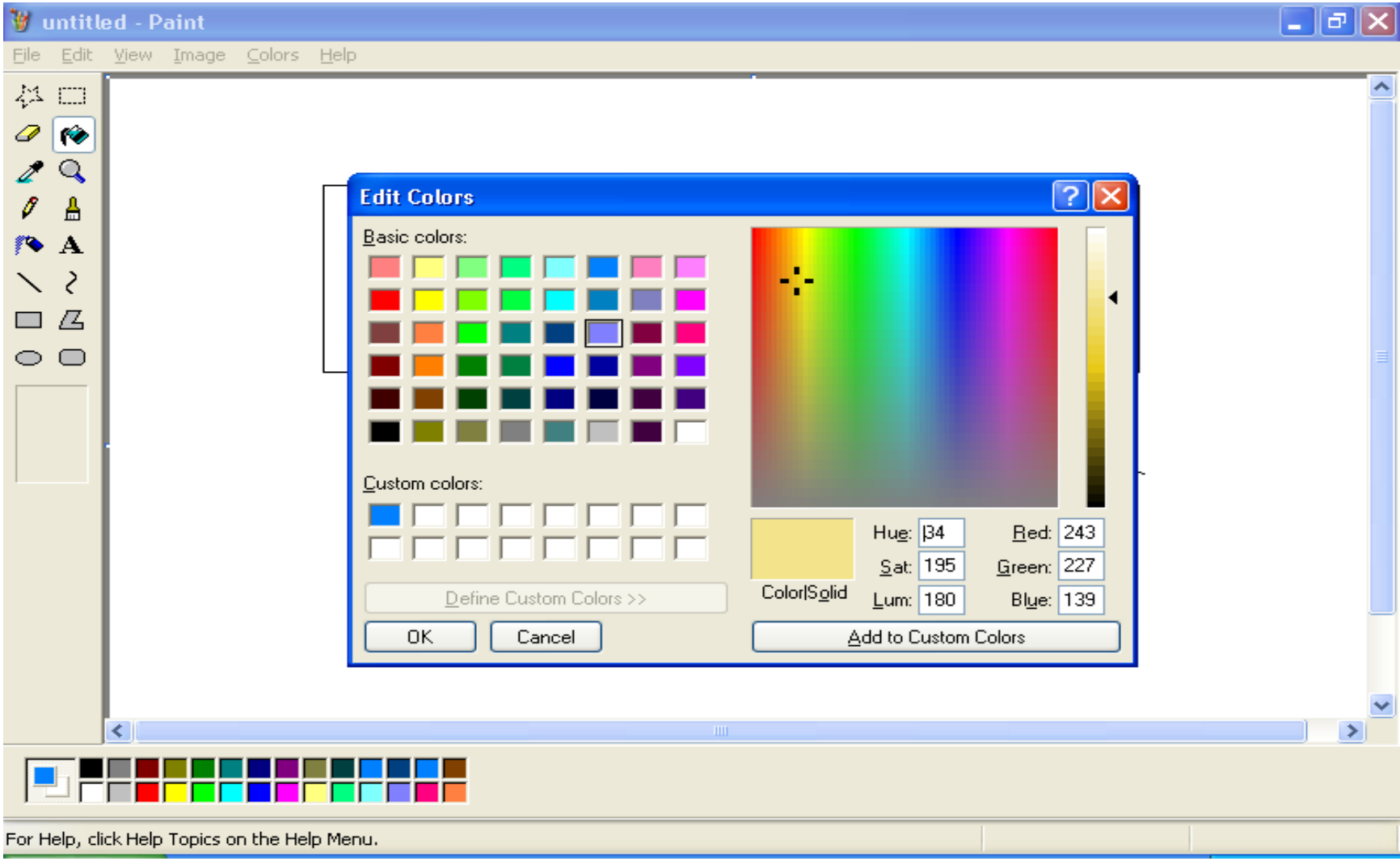


Figure 12-18 **The HLS double cone**


untitled - Paint
File Edit View Image Colors Help
Edit Colors
Basic colors:
Custom colors:
Define Custom Colors >>
OK
Cancel
Color|Solid
Hue: 34
Sat: 195
Lum: 180
Red: 243
Green: 227
Blue: 139
Add to Custom Colors
For Help, click Help Topics on the Help Menu.

Thank You